Inequality

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Narrative arc

Concepts

↓

Individuals

↓

Populations

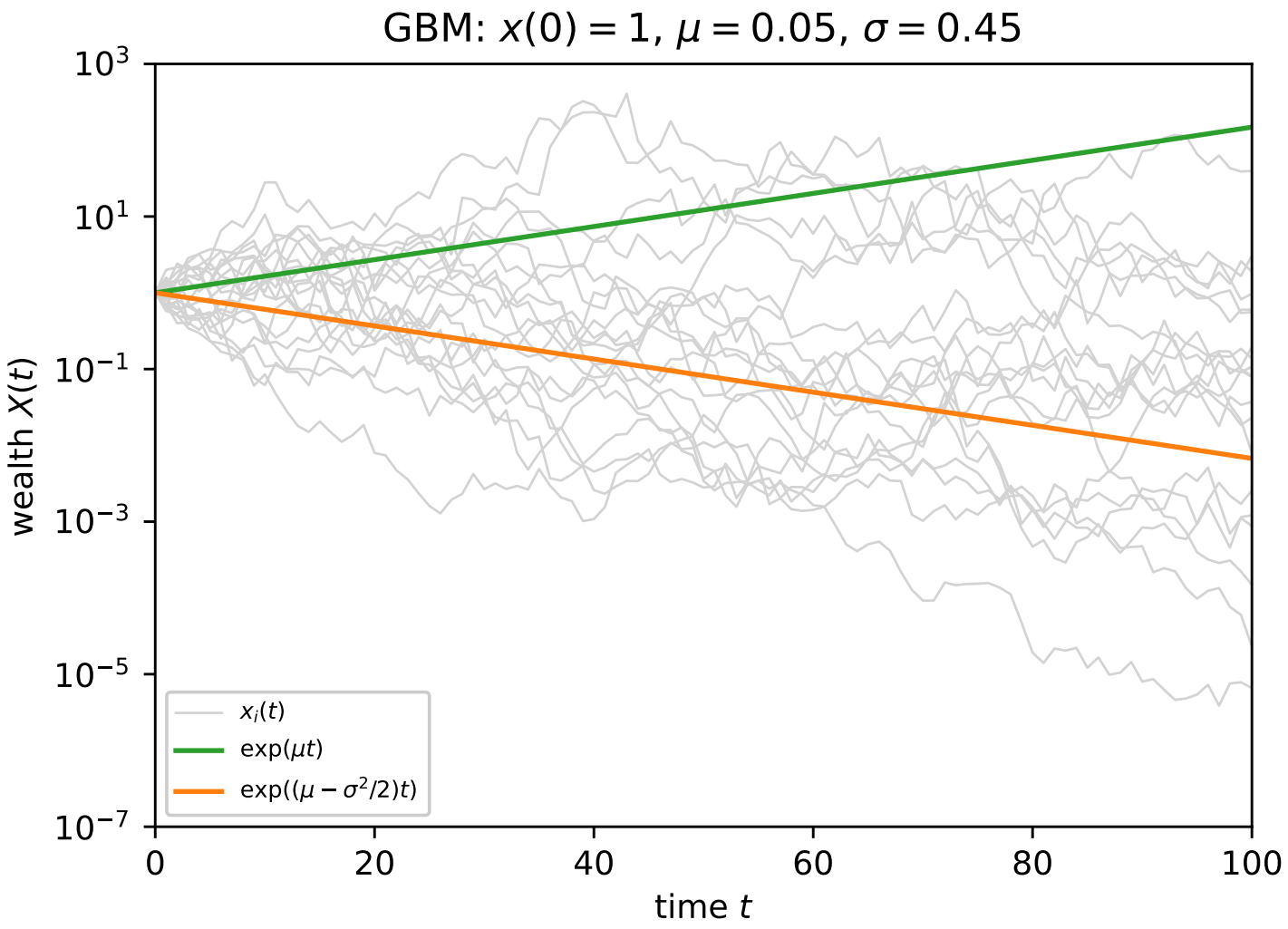
↓

Societies

Aim: set baseline properties before adding interactions.

Model world

N wealths, $x_i(t)$, following independent trajectories of GBM.



Lognormal distribution

8.1.1 Lognormal distribution

Recall from Sec. 4.4 that the SDE for GBM,

$$dx = x(\mu dt + \sigma dW), \tag{8.1}$$

has solution

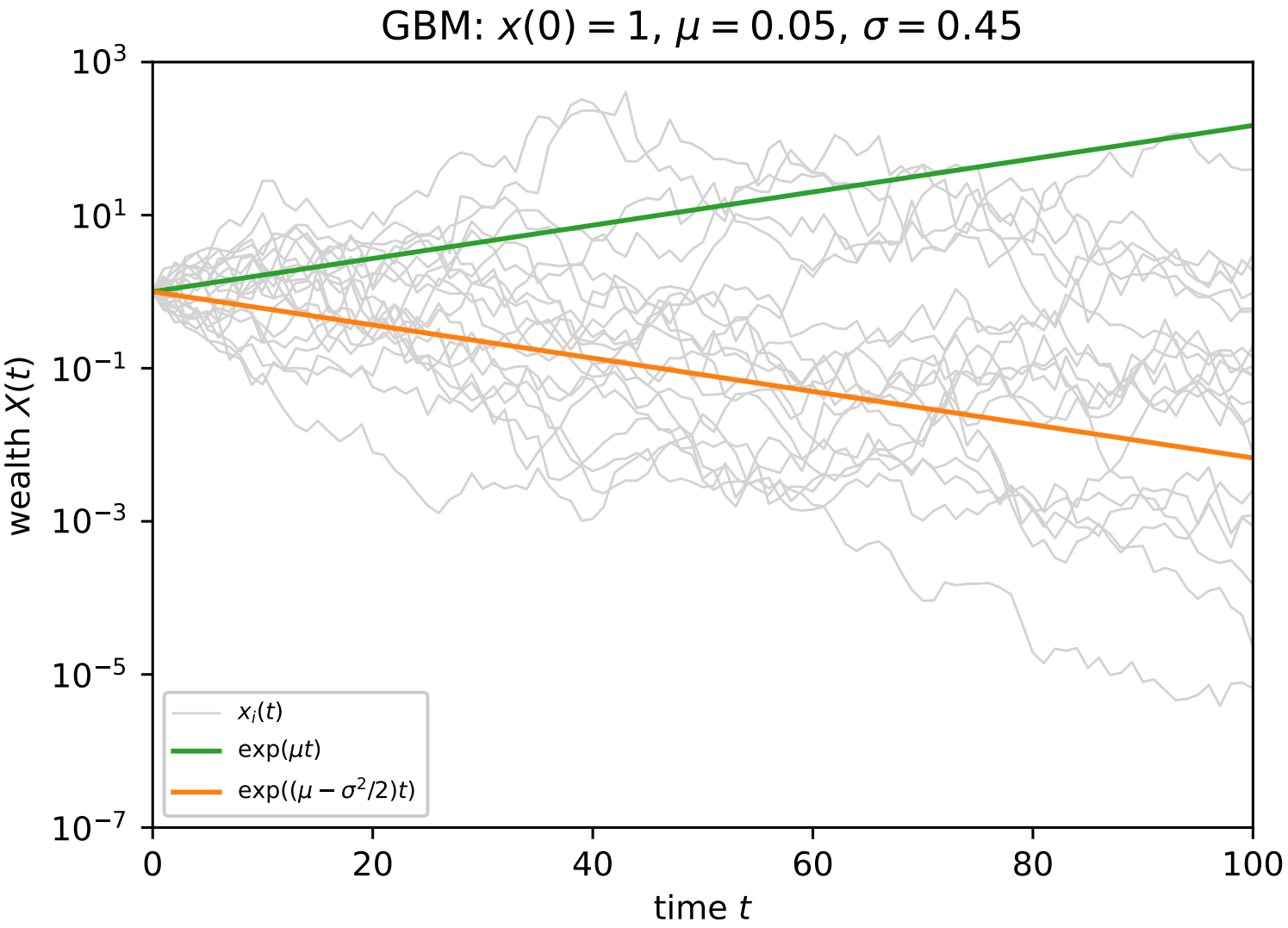
$$x(t) = x(0) \exp\left[\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W(t)\right], \tag{8.2}$$

which, at a given time t, can be viewed as a lognormally-distributed random variable with time-dependent parameters,

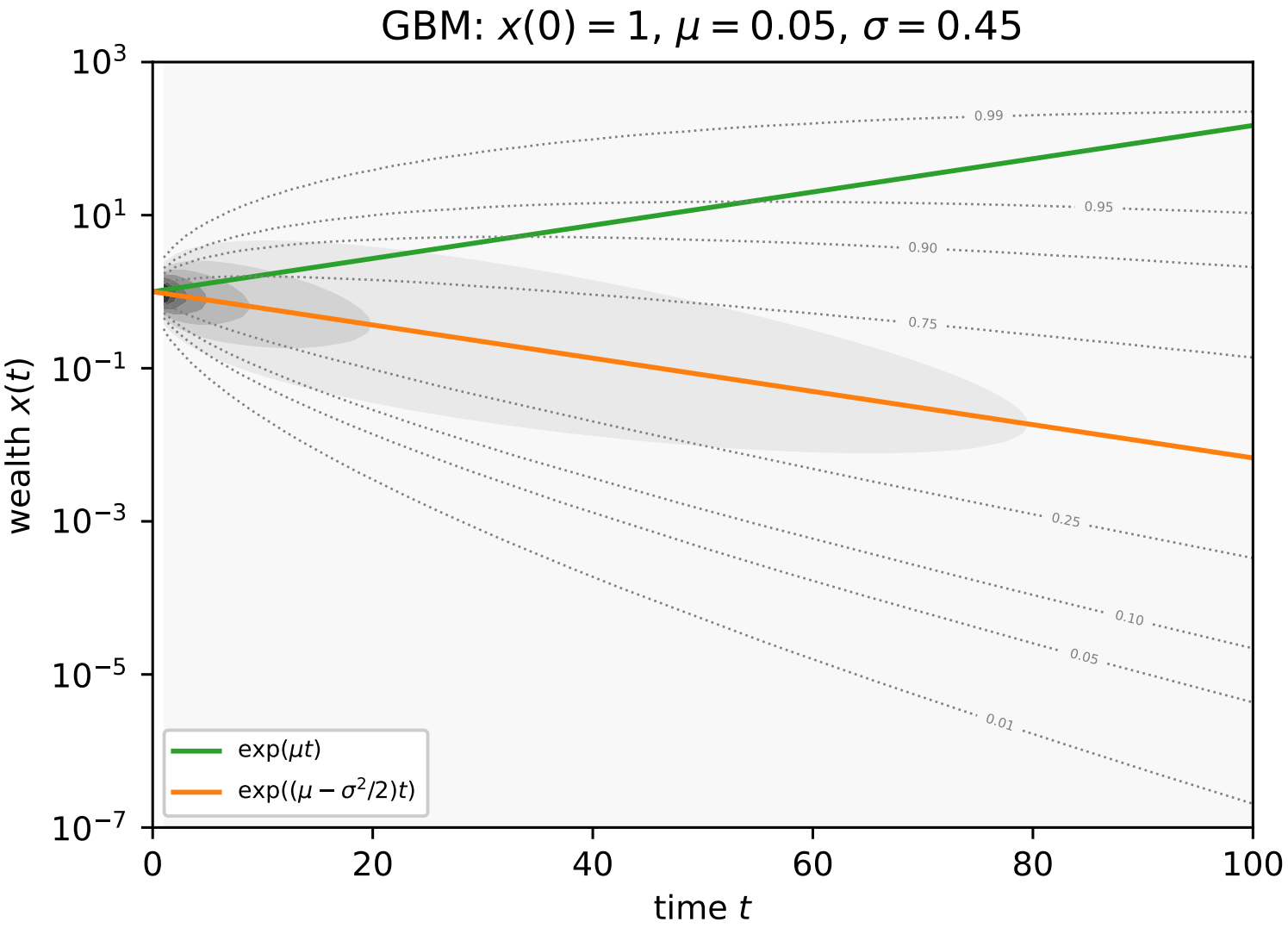
$$\ln x(t) \sim \mathcal{N}\left(\ln x(0) + \left(\mu - \frac{\sigma^2}{2}\right) t, \sigma^2 t\right). \tag{8.3}$$

In our model population, we will assume that all individuals have the same starting wealth, $x(0)$.

Lognormal distribution



Lognormal distribution



Two growth rates

Two ways to quantify "average" wealth growth rate in our model population:

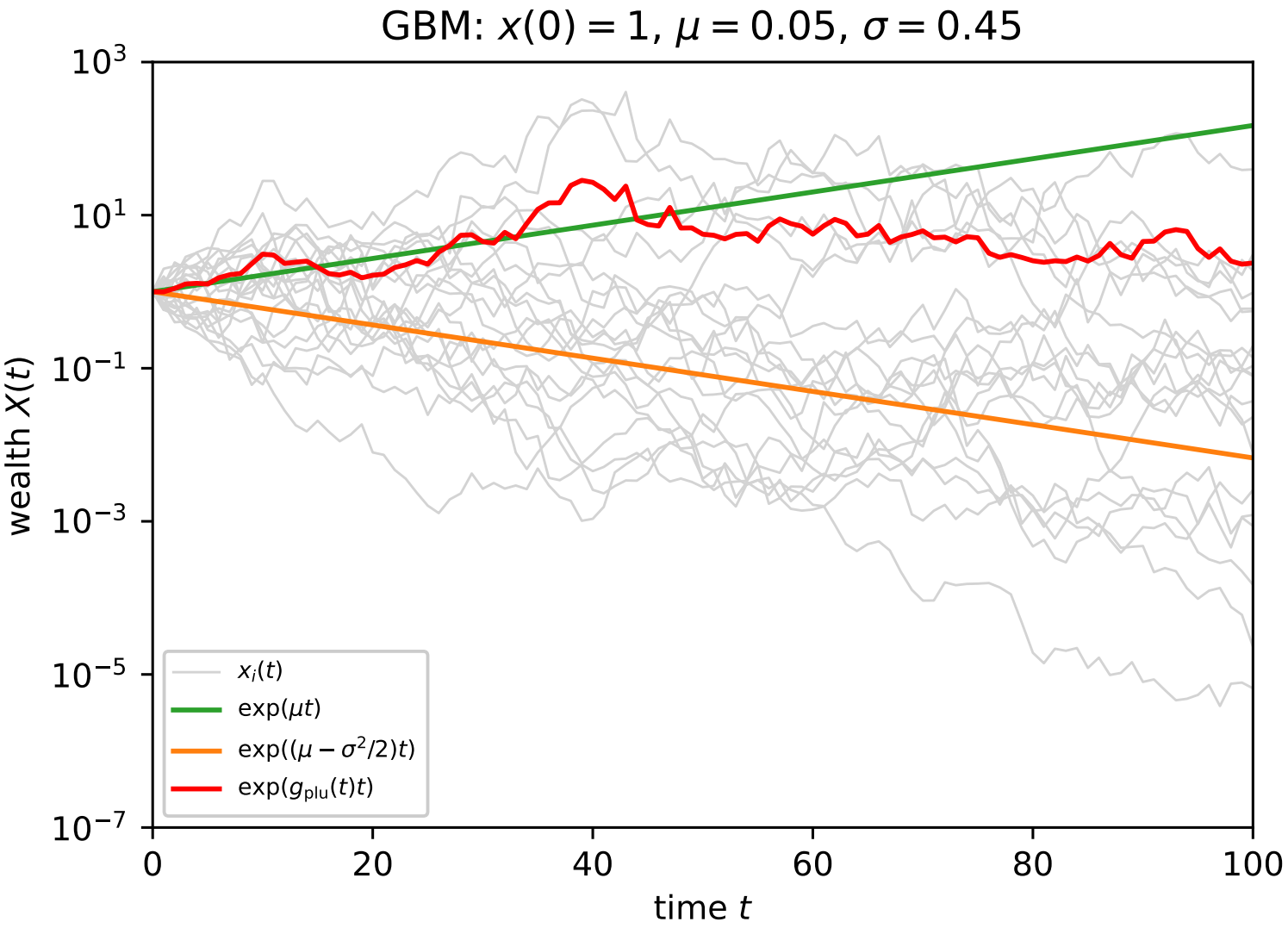
1. Plutocratic growth rate

Find the population average wealth, $\langle x \rangle_N$, and compute its ergodic growth rate:

$$g_{\text{plu}} \equiv \frac{\Delta \ln \langle x \rangle_N}{\Delta t}.$$

"Plutocratic" because each individual makes a wealth-weighted contribution to the average.

Two growth rates



Two growth rates

Two ways to quantify "average" wealth growth rate in our model population:

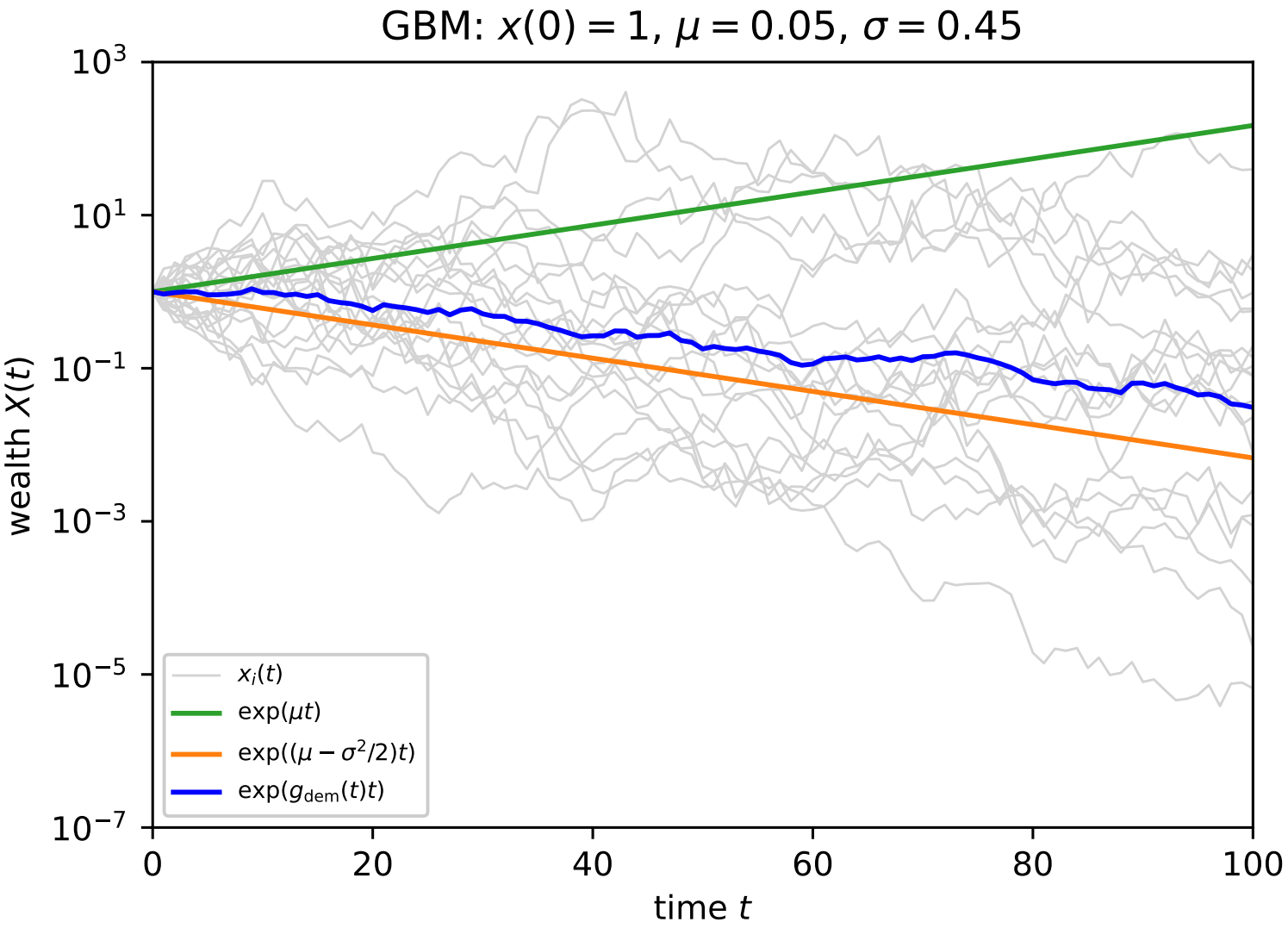
2. Democratic growth rate

Find the ergodic growth rate of each wealth, $\frac{\Delta \ln x_i}{\Delta t}$, and compute the population average:

$$g_{\text{dem}} \equiv \left\langle \frac{\Delta \ln x}{\Delta t} \right\rangle_N .$$

"Democratic" because each individual makes an equal contribution to the average.

Two growth rates



Ergodicity

For finite N, both g_{plu} and g_{dem} are random variables.

We can replace them with model properties as $N \to \infty$:

$$g_{\text{plu}} = \frac{\Delta \ln \langle x \rangle_N}{\Delta t} \quad \to g_{\langle\,\rangle} = \mu;$$

$$g_{\text{dem}} = \left\langle \frac{\Delta \ln x}{\Delta t} \right\rangle_N \quad \to \bar{g} = \mu - \frac{\sigma^2}{2} \text{ (ergodicity).}$$

Good approximations for "large enough" model populations.

Discussion

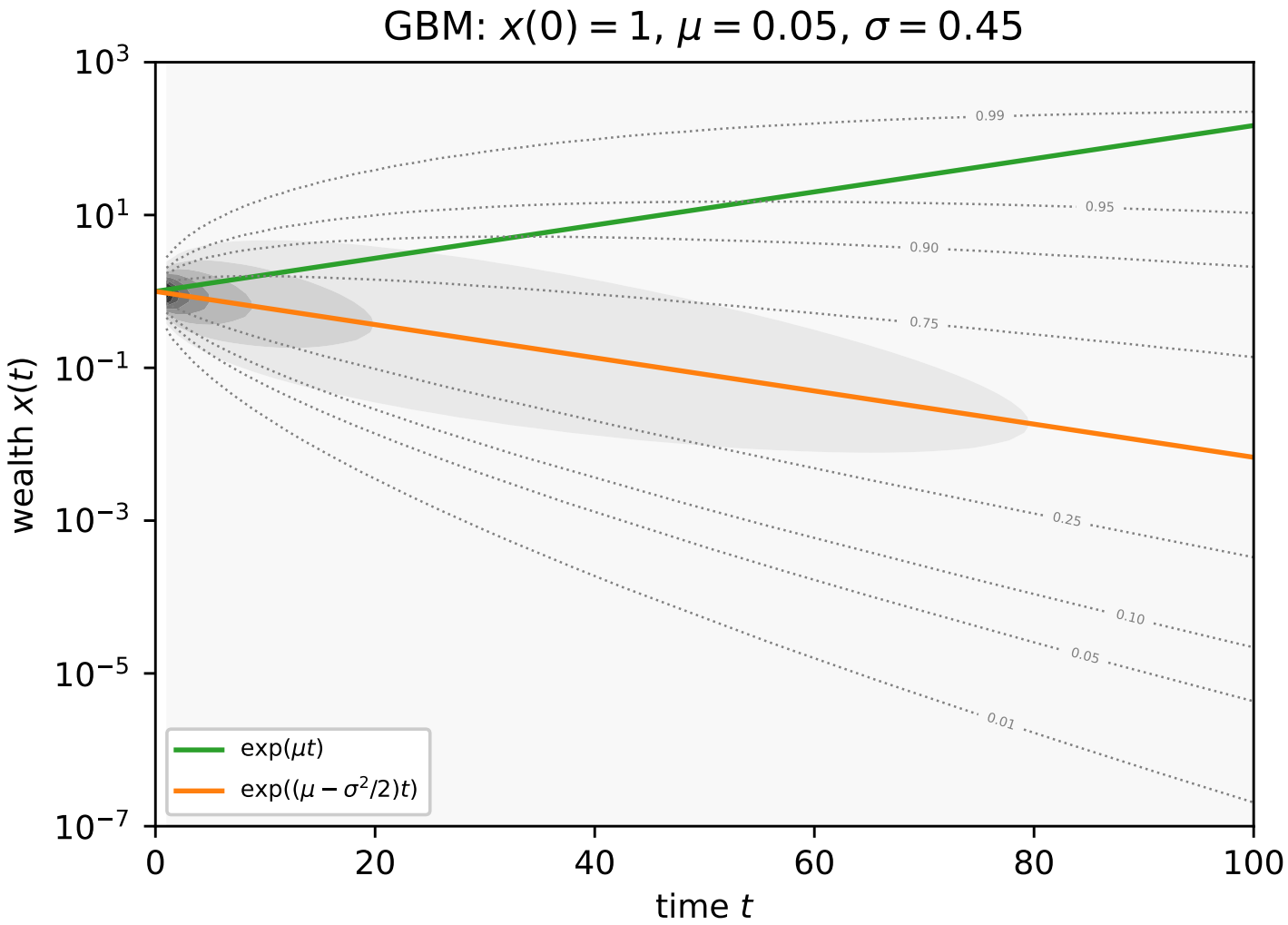
Jeff Bezos in the Library (with the Extreme Wealth).

Gross Domestic Product – government vs. the people?

"The two growth rates of the economy"
Yonatan Berman, OP, AA (2020)
https://arxiv.org/abs/2009.10451.

Inequality

"A whole menagerie of inequality measures in economics."



Recipe for inequality metric à la EE

1. Note that: if $g_{\text{plu}} > g_{\text{dem}}$, then inequality is increasing; and if $g_{\text{plu}} < g_{\text{dem}}$, then inequality is decreasing.

2. Therefore, define inequality, $J(t)$, as the thing that grows at the **difference** between the two rates:

$$\frac{dJ}{dt} = g_{\text{plu}} - g_{\text{dem}}.$$

3. Integrate from time 0 (equality) to t to get:

$$J(t) = \ln \langle x(t) \rangle_N - \langle \ln x(t) \rangle_N .$$

Inequality

J is the **mean logarithmic deviation** or **Theil's L index**.

For a large population, $N \to \infty$, following GBM,

$$\frac{dJ}{dt} = g_{\langle\,\rangle} - \bar{g} = \frac{\sigma^2}{2} \quad \Rightarrow \quad J(t) = \frac{\sigma^2 t}{2},$$

so inequality grows without bound.

For general wealth dynamics with ergodicity transformation, $v(x)$, wealth is v-normally distributed and

$$J(t) = v\left(\langle x(t)\rangle_N\right) - \langle v\left(x(t)\right)\rangle_N .$$

Finite populations

Consider the average wealth, $\langle x(t) \rangle_N$, of our model population.

Tension between two limiting scenarios:

- for large N, $\langle x(t) \rangle_N$ grows at the ensemble-average growth rate, μ ("self-averaging");
- for large t, all trajectories in the population (and therefore $\langle x(t) \rangle_N$) grow at the time-average growth rate, $\mu - \sigma^2/2$.

Question: for a given N, for how long does self-averaging hold?

Finite populations

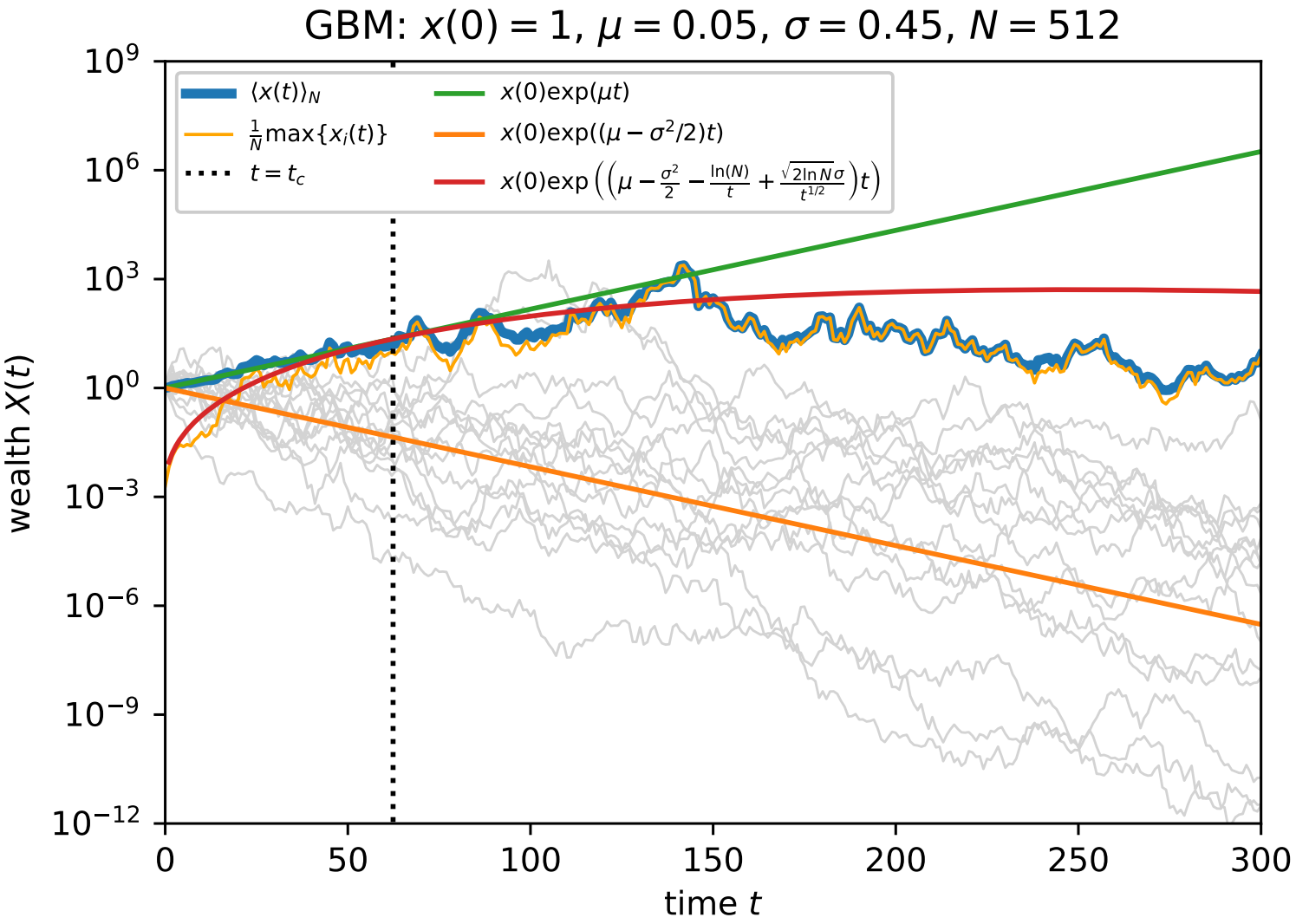
Statistical physics to the rescue – Chapter 8.2.1.

Self-averaging holds when the relative fluctuations in the population average are small (Ginzburg criterion):

$$t < t_c \equiv \frac{2 \ln N}{\sigma^2}.$$

After this, the population average will be dominated by large deviations – and we can predict its evolution.

Finite populations



Thank you.